Armstrong Number: $n = 153 \Rightarrow 1^3 + 5^3 + 3^3 = n \rightarrow$ T / F

power = 0

temp = temp + pow(d, 3)
= 27 + pow(5, 3)
= 27 + 125
temp = 153

if (n == temp)
- true
- false

n%10
d = 153%10 = 3
n = n/10
153/10 = 15
while(n>0) 1/10 = 0
{
power++;
}

Encapsulation: →
The process of wrapping the data members of a class inside a block of code so that they cannot be accidentally modified, is called encapsulation. It is done using the private access modifier.
To provide access outside class the class, we use special methods called "getters" & "setters".
The "getters" & "setters" should be public.

* **Inheritance:** The property of inheriting / accessing the data members of the Parent class into the respective child classes, is called inheritance. Here, the child classes can also have properties of their own & methods of their own.

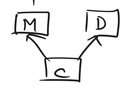
Types of Inheritance:
1. Single
2. Multi-level
3. Multiple
4. Hierarchical
5. Hybrid

} Inheritance

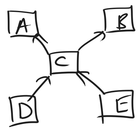
Single Level:

Multi-level:

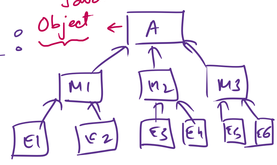
Multiple:



Hybrid:



Hierarchical:



Polymorphism:

Latin
poly ↙ ↘ morph
many — forms / shapes

Anvitha → Classroom : Student
→ Home : Daughter / Sister
→ Restaurant : Customer
→ Playground : Athlete / Player

Anvitha is not changing. Her role is changing.

Types of Polymorphism:

(I) Static Polymorphism : Compile Time : Overloading →
[Same Class]

(II) Dynamic Polymorphism : Runtime : Overriding →
[Multiple Classes]